



Deep0Heat: Operator Learning-based Ultra-fast Thermal Simulation in 3D-IC Design

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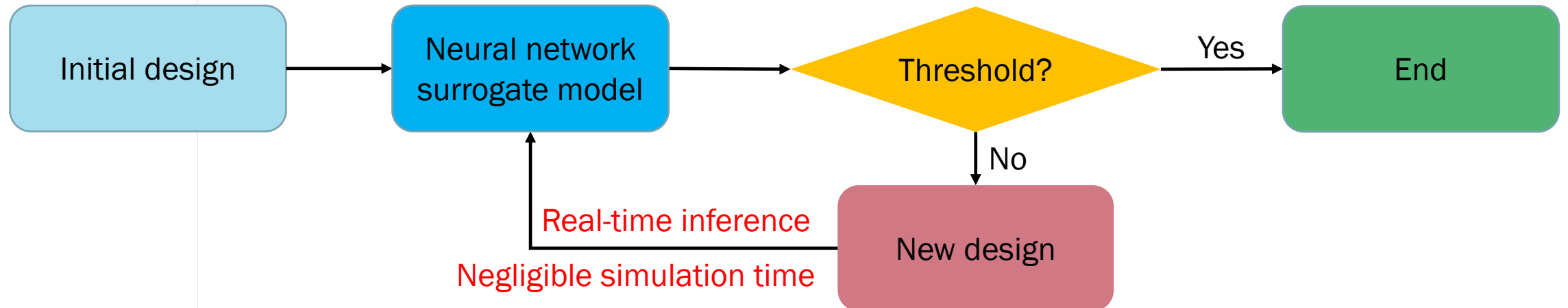
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3D-IC thermal challenges

- 3D IC suffers from thermal issue
 - Increased integration → Higher power density
 - Shorter interconnect lengths → Limited heat dissipation
 - Overall higher temperature reduces reliability and limits performance
- Thermal solver suffers from inefficiency



Motivations

- Limitations of existing NN-based surrogate models
 - Designed for 2D chip or only compatible with 2D power maps ^{[1][2]}
 - Data-driven approaches require massive data collections ^{[2][3][4]}
 - Combined with FEM solver or additional computations ^{[3][4]}
 - Physics-informed NN (PINN) methods ^{[1][5]} don't require data, however
 - Most PINNs are trained for single-case only, some works accept simple parameterizations
 - Still need re-train when non-parametric design configurations change, such as power maps and boundary conditions
 - e.g., Nvidia SimNet™, only takes input from geometric parameters for heat sink design

[1] H. He and J. Pathak, "An unsupervised learning approach to solving heat equations on chip based on auto encoder and image gradient," arXiv preprint arXiv:2007.09684, 2020.

[2] R. Ranade, H. He, J. Pathak, N. Chang, A. Kumar, and J. Wen, "A thermal machine learning solver for chip simulation," in ACM/IEEE Workshop on Machine Learning for CAD, 2022, pp. 111–117.

[3] J. Wen, S. Pan, N. Chang, W.-T. Chuang, W. Xia, D. Zhu, A. Kumar, E.-C. Yang, K. Srinivasan, and Y.-S. Li, "Dnn-based fast static on-chip thermal solver," in Semiconductor Thermal Measurement, Modeling & Management Symposium, 2020, pp. 65–75.

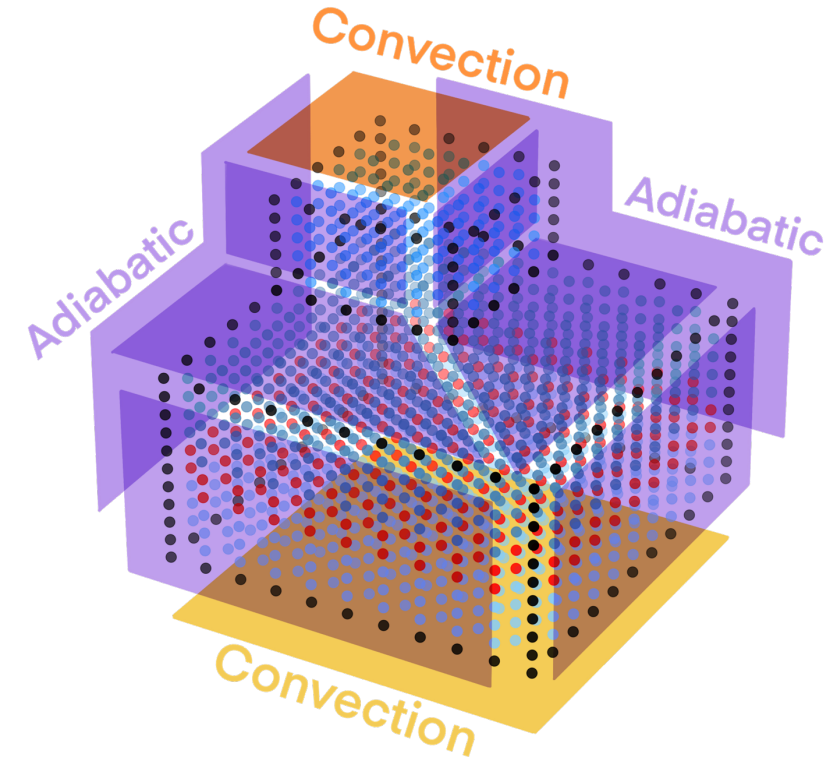
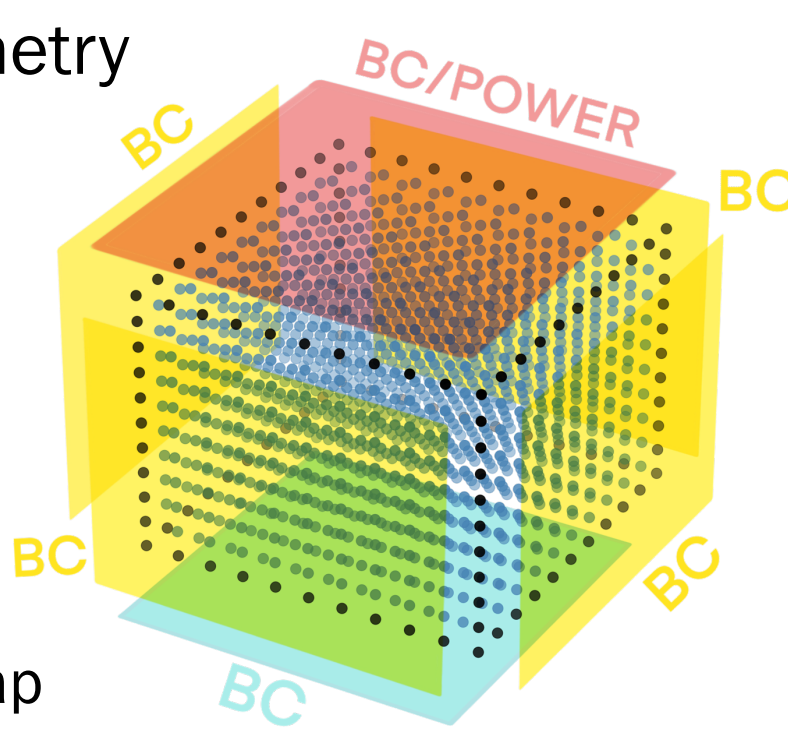
[4] A. Kumar, N. Chang, D. Geb, H. He, S. Pan, J. Wen, S. Asgari, M. Abarham, and C. Ortiz, "ML-based fast on-chip transient thermal simulation for heterogeneous 2.5 d/3D IC designs," in International Symposium on VLSI Design, Automation and Test, 2022, pp. 1–8.

[5] O. Hennigh, S. Narasimhan, M. A. Nabian, A. Subramaniam, K. Tangsali, Z. Fang, M. Rietmann, W. Byeon, and S. Choudhry, "Nvidia simnet™: An ai-accelerated multi-physics simulation framework," in Int. Conf. Computational Science. Springer, 2021, pp. 447–461.



Modular chip configurations

- Stacked cuboidal geometry
- Boundary conditions
 - Dirichlet
 - Neumann (Adiabatic)
 - Robin (Convection)
- Power maps
 - 2D surface power map
 - 3D volumetric power map
- Full-chip material conductivity



Operator learning

- Learn a mapping between function spaces

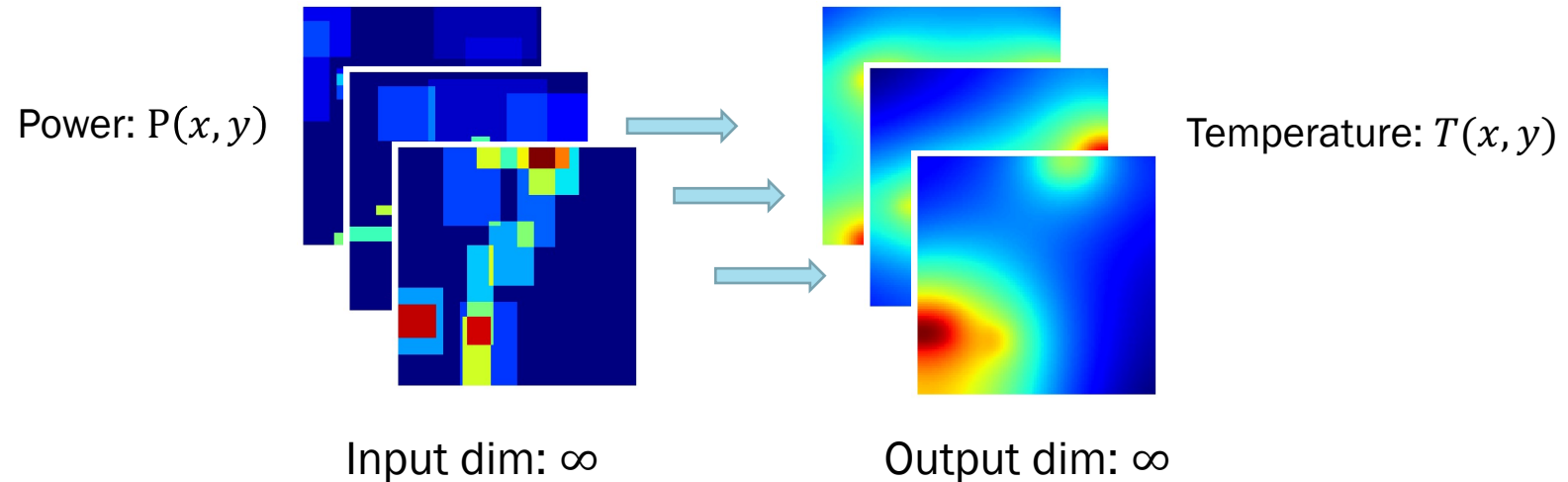
Function: $\mathbb{R}^{d_1} \rightarrow \mathbb{R}^{d_2}$
e.g., image classification



Input dim: number of pixels
Output dim: number of classes

Function: mapping between finite dimensional spaces

Operator: function $\rightarrow$ function
e.g., derivative: $x(t) \rightarrow x'(t)$
e.g., integral: $x(t) \rightarrow \int K(s, t)x(s)ds$
e.g., thermal simulation: $P(x, y) \rightarrow T(x, y)$

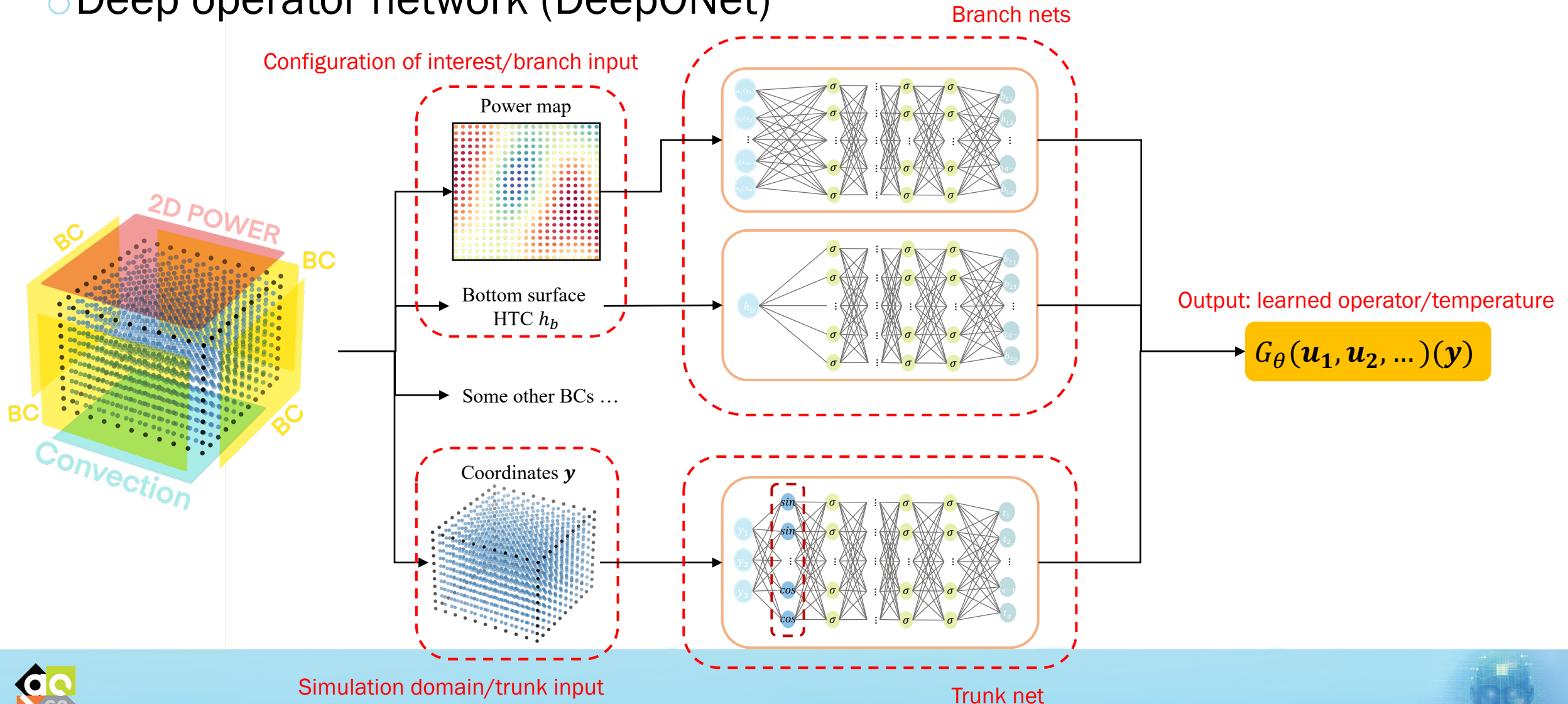


Operator: Mapping between infinite dimensional spaces



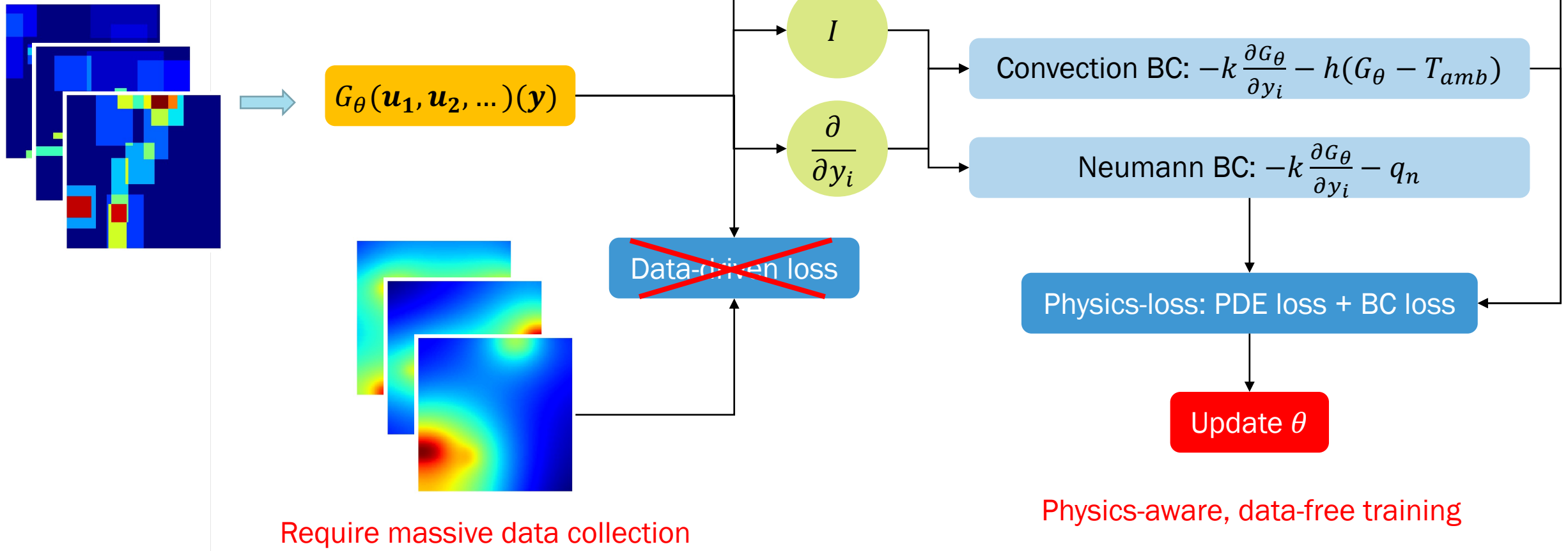
Deep0Heat framework

- Deep operator network (DeepONet)



Deep0Heat framework

Physics-informed data-free training



Case 1: 2D power map

- Problem setting:

- 21*21*11 mesh grid-based single-cuboid geometry
- Define an **arbitrary** 2D surface power map on top
- Adiabatic BC on all side surfaces
- Convection BC on bottom with $HTC = 500 \text{ W}/(\text{m}^2\text{K})$ and $T_{\text{amb}} = 298.15 \text{ (K)}$

- Training settings:

- Generate random 2D power maps from a Gaussian random field with zero mean and one standard deviation, in which length scale parameter equals to 0.3
- 10 hours training on a single Tesla V100 GPU

- Test settings:

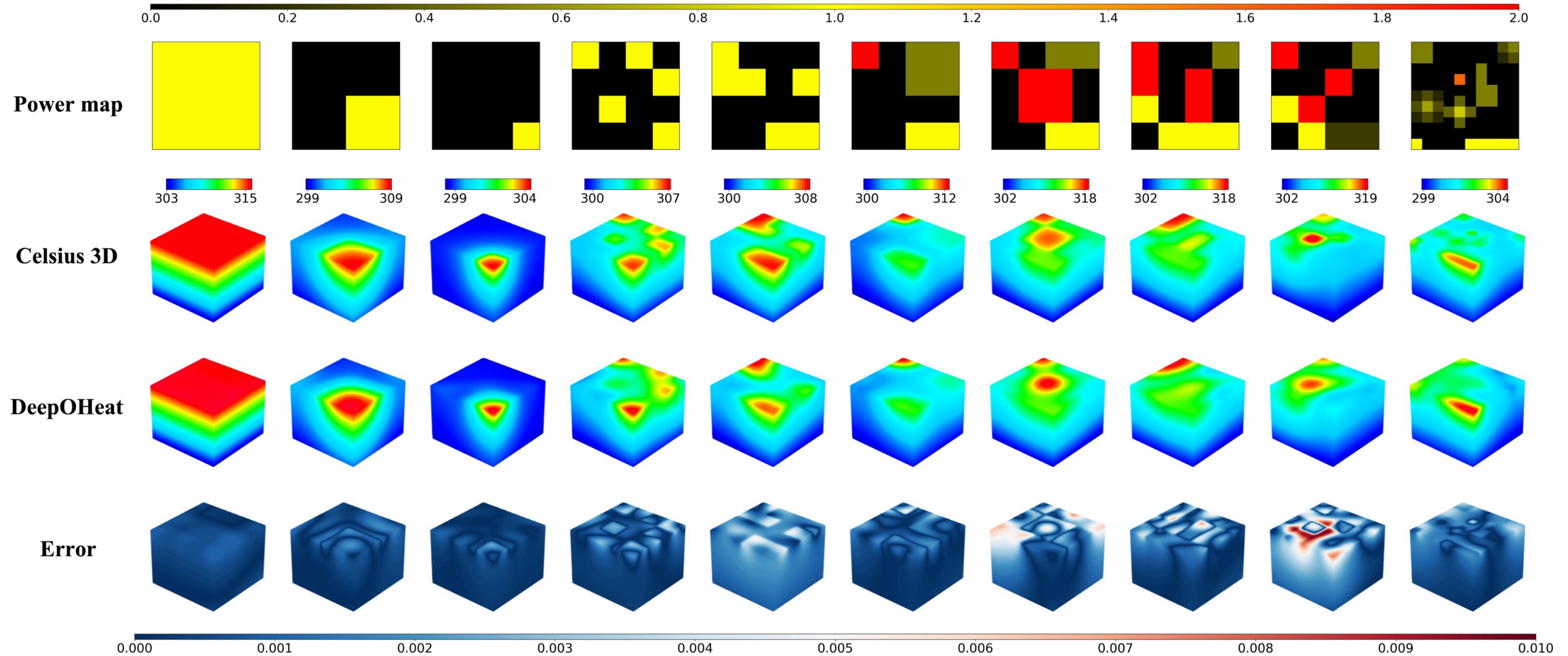
- Discrete tile-based power maps, compared against Celsius 3D from Cadence



Case 1: 2D power map

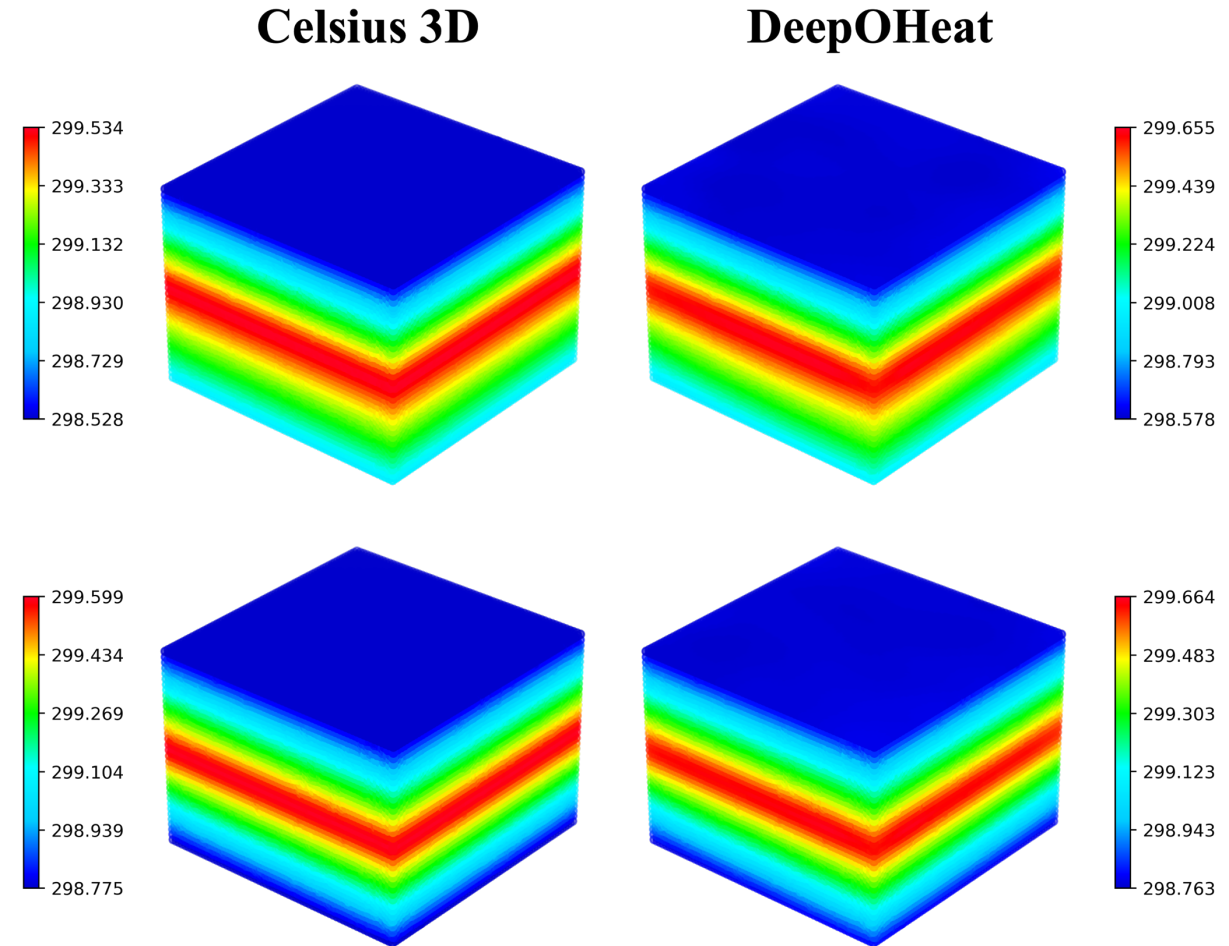
	p_1	p_2	p_3	p_4	p_5	p_6	p_7	p_8	p_9	p_{10}
MAPE (%)	0.03	0.03	0.02	0.05	0.14	0.04	0.13	0.07	0.16	0.08
PAPE (%)	0.10	0.20	0.24	0.38	0.52	0.49	0.71	0.66	1.00	0.40

Celsius: 5 minutes (CPU); DeepOHeat: 0.1 s (CPU) → **3000x** speed up, 0.001 s (GPU) → **300000x** speed up



Case 2: HTC on top and bottom

- Problem settings:
 - Volumetric power
 - Changeable HTC on top and bottom
 - Range: [333.33, 1000]
- Training: 2 hours
- Results:
 - MAPE: 0.032%, 0.011%
 - PAPE: 0.043%, 0.025%
 - Celsius: 2 minutes
 - Speed up: **1200x** (CPU), **120000x** (GPU)



Conclusion & Discussion

○ Conclusion

- Deep0Heat is the first physics-aware operator learning framework for ultra-fast 3D chip thermal simulation
- Deep0Heat achieves satisfactory accuracy on unseen new power maps with negligible computation time when compared against Cadence Celsius 3D

○ Discussion & Future work

- 3D power map and conductivity as changeable configurations
- Training cost and difficulty for high dimensional input

